

GPU scale-out does not remove a shared media-preparation bottleneck

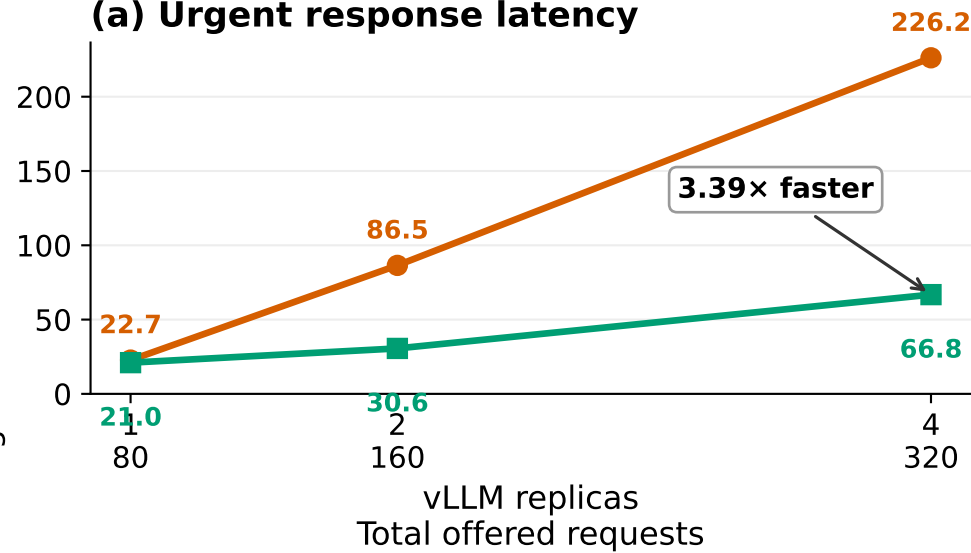
Offered load scales with replicas; the shared CPU preparation pool remains fixed at 4 workers

Engine-only priority
(FCFS media preparation)

End-to-end priority

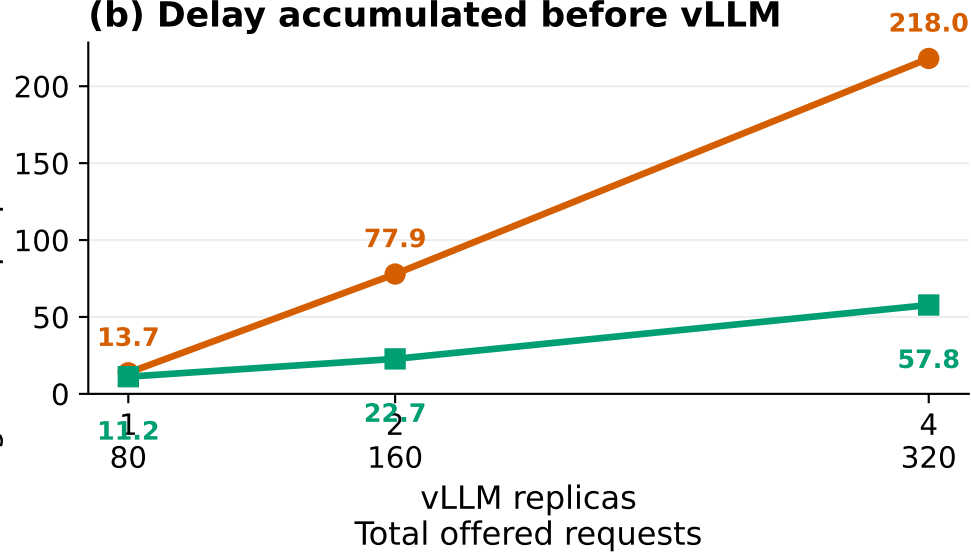
Urgent mean end-to-end TTFT (s)

(a) Urgent response latency



Urgent mean preparation wait (s)

(b) Delay accumulated before vLLM



Controlled matched workload; one run per point. All configurations use native vLLM priority.